Methods 7

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1.

a.

The Fisher sharp null hypothesis assumes that class size makes exactly no difference in the outcome. In other words, it assumes that each class would have gotten the exact same result regardless of class size, small or regular.

b.

From Table 9.1:

School/Stratum	No. of Classes	Regular Classes ($W_i = 0$)	Small Classes ($W_i = 1$)
1	4	$-0.197, 0.236$	$0.165, 0.321$
2	4	$0.117, 1.190$	$0.918, -0.202$
3	5	$-0.496, 0.225$	$0.341, 0.561, -0.059$
4	4	$-1.104, -0.956$	$-0.024, -0.450$
5	4	$-0.126, 0.106$	$-0.258, -0.083$
6	4	$-0.597, -0.495$	$1.151, 0.707$
7	4	$0.685, 0.270$	$0.077, 0.371$
8	6	$-0.934, -0.633$	$-0.870, -0.496, -0.444, 0.392$
9	4	$-0.891, -0.856$	$-0.568, -1.189$
10	4	$-0.473, -0.807$	$-0.727, -0.580$
11	4	$-0.383, 0.313$	$-0.533, 0.458$
12	5	$0.474, 0.140$	$1.001, 0.102, 0.484$
13	4	$0.205, 0.296$	$0.855, 0.509$
14	4	$0.742, 0.175$	$0.618, 0.978$
15	4	$-0.434, -0.293$	$-0.545, 0.234$
16	4	$0.355, -0.130$	$-0.240, -0.150$

To make this table as an object in R,

```
library(tidyverse)
library(survey)

# elongate table so each row is a class
## (so we don't have to deal with columns w/ vectors)

school = c(rep(1, 2),rep(2, 2),rep(3, 2),rep(4, 2),rep(5, 2),rep(6, 2),
           rep(7, 2),rep(8, 2),rep(9, 2),rep(10, 2),rep(11, 2), rep(12, 2),
           rep(13, 2), rep(14, 2),rep(15, 2), rep(16, 2), rep(1, 2),rep(2, 2),
           rep(3, 3),rep(4, 2),rep(5, 2),rep(6, 2),rep(7, 2),rep(8, 4),
           rep(9, 2),rep(10, 2),rep(11, 2), rep(12, 3),
           rep(13, 2), rep(14, 2),rep(15, 2), rep(16, 2))

w_i = c(rep(0, 32), rep(1, 36))


reg = c(-0.197, 0.236, 0.117,1.190,-0.496,0.225,-1.104,-0.956,-0.126,
        0.106,-0.597,-0.495,0.685,0.270,-0.934,-0.633,-0.891,-0.856,
        -0.473,-0.807,-0.383,0.313,0.474,0.140,0.205,0.296,0.742,0.175,
        -0.434,-0.293,0.355,-0.130)

small = c(0.165, 0.321, 0.918, -0.202, 0.341, 0.561, -0.059, -0.024,
          -0.450, -0.258, -0.083, 1.151, 0.707, 0.077, 0.371, -0.870,
          -0.496, -0.444, 0.392, -0.568, -1.189, -0.727, -0.580, -0.533,
          0.458, 1.001, 0.102, 0.484, 0.855, 0.509, 0.618, 0.978,
          -0.545, 0.234, -0.240, -0.150)

y = c(reg, small)

tps = tibble(
  school = school,
  w_i = w_i,
  y = y
)
```

From Section 9.4.2, denoting our number of strata as J, then for some J-component vector λ, we have statistic

$$T^{dif,\lambda} = \left| \sum_{j=1}^{J} \lambda(j) \cdot \left(\bar{Y}_t^{obs}(j) - \bar{Y}_c^{obs}(j) \right) \right|; \quad \lambda(j) = \frac{N(j)}{N}$$

$$T^{dif,\lambda_{RSS}} = \left| \sum_{j=1}^{J} \frac{N(j)}{N} \cdot \left(\bar{Y}_t^{obs}(j) - \bar{Y}_c^{obs}(j) \right) \right|$$

To compute for the observed data, using the following formula and code:

$$T^{dif,\lambda_{RSS}} = \left| \sum_{j=1}^{16} \frac{N(j)}{N} \cdot \left(\bar{Y}_t^{obs}(j) - \bar{Y}_c^{obs}(j) \right) \right|$$

```
n_class = tps |> nrow()

T_obs = tps |>
  group_by(school) |>
  summarize(
    y_t = sum(w_i * y),
    y_c = sum((1-w_i) * y),
    n_j = n()
  ) |>
  mutate(T_j = (n_j/n_class) * (y_t - y_c)) |>
  summarize(abs(sum(T_j))) |>
  pull()

paste0("The observed data gave a test statistic of approximately ", round(T_obs, 4))
```

```
[1] "The observed data gave a test statistic of approximately 0.4528"
```

c.

```
set.seed(10)

tps_rand = tps

rand_1 = tps |> filter(school == 16) |> select(w_i) |> pull() |> sample() #randomization
indices_16 = which(tps$school == 16)
for (i in 1:length(indices_16)) {
  new_w = rand_1[i]
  tps_rand$w_i[indices_16[i]] = new_w
}
```

```r
n_class = tps_rand |> nrow()

T_rand = tps_rand |>
  group_by(school) |>
  summarize(
    y_t = sum(w_i * y),
    y_c = sum((1-w_i) * y),
    n_j = n()
  ) |>
  mutate(T_j = (n_j/n_class) * (y_t - y_c)) |>
  summarize(abs(sum(T_j))) |>
  pull()

paste0("The data after randomizing the 16th school's assignment groups gave a test statistic
```

```
[1] "The data after randomizing the 16th school's assignment groups gave a test statistic of
```

d.